Khadijah Aamir Chaudhry

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LinkedIn · GitHub · Portfolio

EDUCATION

Lahore University of Management Sciences (LUMS) *2023–2027*
BSc Computer Science · Minor in Computer Engineering

Lahore Grammar School *2018–2023*
A Levels: 4 As · O Levels: 9 A*s · SAT: 1530

RESEARCH INTERESTS

Secure embedded systems · Hardware roots of trust · Hardware–software co-design · Trusted execution · Applied cryptography

RESEARCH EXPERIENCE

Research Assistant, CHASE Lab, LUMS *Aug 2025–present*
Supervisor: Dr. Muhammad Ali Siddiqi

- **Built a simulated Rocket Chip and Verilator testbed** to capture RISC-V transaction traces for research on runtime hardware Trojan detection.
- **Benchmarked three post-quantum algorithms** on ESP32, comparing latency, memory, network cost, and estimated energy on constrained hardware.

Research Assistant, SysNet Lab, LUMS *Jan–May 2026*
Supervisor: Dr. Naveed Anwar Bhatti

- **Co-developed EyeSense's smartphone ToF and machine-learning pipeline** to explore accessible refractive-error screening.
- **Supported lab and eye-camp data collection** for a 120-participant team study that reported 1.15 D mean absolute error.

SELECTED PROJECTS

On Chip Behavioral Monitoring · early draft *2026–present*

- **Configured four RV32 Rocket Chip variants** and captured TileLink/MMIO traces with Verilator for runtime monitoring research.
- **Replayed benign and trigger-active firmware** to create traces for later detector evaluation.

Protected Media Containers · early draft *2026–present*

- **Researched a trusted-execution design** that keeps camera and audio plaintext protected from an untrusted mobile operating system.
- **Specified encrypted storage, display, and sharing flows** to preserve user policy controls on an untrusted phone.

EyeSense: Mobile Eyesight Screening · project write-up *Jan–May 2026*

- **Co-developed a phone-based ToF and ML pipeline** to explore lower-cost refractive-error screening using phase-map changes.
- **Evaluated the method with a 120-participant team** that reported 1.15 D mean absolute error for preliminary eyesight screening.

Post-Quantum Cryptography on ESP32 · project write-up · code *Aug–Dec 2025*

- **Implemented and benchmarked three post-quantum algorithms** on ESP32 to assess cryptography on small devices.
- **Compared latency, memory, network cost, and estimated energy** to identify practical trade-offs for embedded deployment.

AasPass Peer Tutoring · app demo

- **Led Android authentication and administrator features** for a student–tutor app, including LUMS email verification and profile setup.

- **Added moderation controls** to help administrators manage the peer tutoring community.

Spotify Mood Classification and Recommendation · write-up

- **Analyzed audio features with PCA and K-means** to explore how song characteristics relate to listening moods.
- **Built mood classification and similarity recommendations** to help listeners find music with a similar feel.

ROM-Based Digital Lock · project report

- **Implemented an EPROM keypad lock** with state-machine logic as a physical access-control demonstration.
- **Added an unlock indicator and incorrect-entry alarm** to make accepted and rejected codes clear to users.

TEACHING EXPERIENCE

Teaching Assistant, CS100, LUMS *Aug 2026–present*

- **Designed and conducted programming labs** and prepared and graded quizzes for about 450 first-year students.
- **Held weekly office hours and invigilated exams** to provide students with regular help and assessment support.

PROFESSIONAL EXPERIENCE

Marketing Manager, Fintelligent Solutions *Jul 2026–present*

- **Coordinated delivery of the brand identity and office collateral** to establish a consistent public identity for the consultancy.
- **Coordinated LinkedIn launch content and the company website** to help prospective clients understand services and make contact.

Data Collection Coach, Uplift AI *Jul–Aug 2024*

- **Led 10 data collectors and organized field outreach** for a Pakistani-language AI dataset.
- **Ran daily quality reviews and feedback sessions** to improve consistency in collected examples.

LEADERSHIP & SERVICE

Peer Ambassador Leader, LUMS *Aug 2026–present*

- **Led orientation and faculty introductions** to help 15 incoming students settle into university.
- **Continue monthly support sessions** to provide students with a familiar point of contact through their first year.

Women in STEM Coordinator, LUMS SPADES *Sep 2024–May 2025*

- **Helped launch SPADES' first women-only STEM event** at PSIFI to welcome more women into STEM programming.
- **Welcomed more than 100 participants** to a dedicated science and technology program.

Co-Founder and Logistics Director, Umeed-e-Nau NGO *Sep 2021–May 2023*

- **Co-founded an NGO** to create inclusive learning opportunities for underprivileged and disabled children.
- **Organized educational activities and visits** that helped children learn and participate together.

SELECTED GRADUATE COURSEWORK

Internet of Things · Applied Cryptography · Topics in Computer and Network Security · Distributed Systems

HONORS AND AWARDS

- **100% Merit Scholarship**, Lahore Grammar School · *Aug 2021–May 2023*
- **Shortlisted writing submission**, LUMSx Nexus · *May 2025*
- **Honorable Mention**, EducationUSA Pakistan Writing Contest · *Sep 2022*

TECHNICAL SKILLS

Programming languages: C · C++ · Python · JavaScript
Embedded and hardware: ESP32 · RISC-V · Rocket Chip · Verilator · FreeRTOS
Machine learning and sensing: ToF sensing · CNNs · ridge regression
Tools and methods: Docker · Git · post-quantum cryptography benchmarking